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TAIWANESE NORMS FOR THE HARVARD GROUP SCALE OF HYPNOTIC SUSCEPTIBILITY, FORM A

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Abstract: Hypnotic susceptibility is a fundamental individual characteristic to consider in studies examining hypnosis. Although there is no existing normative data of group hypnotic susceptibility tests for the Mandarin-speaking Chinese population, the current study administered the Mandarin Chinese translation of the Harvard Group Scale of Hypnotic Susceptibility, Form A (HGSHS:A) to 242 subjects (137 females and 105 males). The results indicate that the normative properties—including the score distribution, item pass rates, and reliability—are comparable to 15 reference samples. In general, the Mandarin Chinese version of the HGSHS:A can be used as a viable and reliable instrument for prescreening subjects' hypnotizability in the Mandarin Chinese-speaking population in Taiwan.

Hypnosis refers to an altered state of consciousness that is elicited by a hypnotic induction and a specific suggestion. In this state of altered consciousness, the subject can have focused attention, reduced peripheral awareness, and even enhanced capacity to respond to suggestion (Elkins, Barabasz, Council, & Spiegel, 2015; Kihlstrom, 1992; Oakley & Halligan, 2013). Hypnotic suggestion has been applied as a tool to study vision (Bryant & McConkey, 1999; Kosslyn, Thompson, Costantini-Ferrando, Alpert, & Spiegel, 2000), attention (Egner, Jamieson, & Gruzelier, 2005; Iani, Ricci, Gherri, & Rubichi, 2006; Raz, Kirsch, Pollard, & Nitkin-Kaner, 2006; Raz, Shapiro, Fan, & Posner, 2002), memory (Barnier, 2002; Barnier, McConkey, & Wright, 2004; Cox & Barnier, 2003; Mendelsohn, Chalamish, Solomonovich, & Dudai, 2008), perception (Nash, Lynn, & Stanley, 1984; Spiegel, Bierre, & Rootenberg, 1989), motor control (Blakemore, Oakley, & Frith, 2003; Haggard, Cartledge, Dafydd, & Oakley, 2004), pain (Derbyshire, Whalley, & Oakley, 2009; Derbyshire, Whalley, Stenger, &

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Oakley, 2004; Hofbauer, Rainville, Duncan, & Bushnell, 2001; Rainville, Duncan, Price, Carrier, & Bushnell, 1997; Rainville et al., 1999), and hallucinations (Bryant & Mallard, 2003; Mazzoni et al., 2009; Szechtman, Woody, Bowers, & Nahmias, 1998).

To adopt hypnotic suggestion as a tool in studying cognitive processes, it is essential to ensure the effectiveness of the induction procedure on individual subjects. The effectiveness of hypnotic suggestion highly depends on the subject's hypnotizability, which is often regarded as a personality variable (Hilgard, 1965). Since the 1950s, there have been studies trying to establish standard scales for reliable assessment of hypnotizability. The first was the Stanford Hypnotic Susceptibility Scale, Forms A and B (SHSS:A & B; Weitzenhoffer & Hilgard, 1959), which has mostly been supplanted by its more modern successor (Barnier & McConkey, 2004; Perry, Nadon, & Button, 1992), the Stanford Hypnotic Susceptibility Scale, Form C (SHSCC:C; Weitzenhoffer & Hilgard, 1962). Notwithstanding that the SHSS:C has been widely adopted to measure hypnotizability, the Harvard Group Scale of Hypnotic Susceptibility, Form A (HGSHS:A; Shor & Orne, 1962) is likely the most frequently used group adaptation of the SHSS:A & B for assessing hypnotizability (Barnier & McConkey, 2004). In contrast to the SHSS:C's individual assessment approach, the HGSHS:A is not just a less resource-intensive instrument for evaluating hypnotizability but can also be administered in a standardized group condition (Benham, Smith, & Nash, 2002; Carvalho, 2013; Költő, Gősi-Greguss, Varga, & Bányai, 2015). It is worth noting that Bentler and Hilgard (1963) found no significant differences in hypnotizability scores either between testing in individual and group settings, or between small (four to six subjects) and large (39 to 52 subjects) groups. Therefore, the HGSHS:A has been regarded as a reliable and efficient tool to screen hypnotizability for large groups in one single session (Benham et al., 2002; Oakley, 2006).

The sample sizes of the HGSHS:A group sessions vary from four to 250. Currently for the Asian population, there is only one peer-reviewed publication on the HGSHS:A, which did not provide normative information such as grade distribution or item difficulty. Hence, the current study aims to establish a normative sample for the Mandarin Chinese-speaking population in Taiwan. In order to use hypnosis as a tool for studying cognitive processing, both locally and cross-culturally, the current study presents Taiwanese data acquired with the Mandarin Chinese translation of the HGSHS:A and compares it to data reported by 15 other reference samples (see Table 1). Moreover, as some studies have suggested that hypnotizability is likely related to gender (Page & Green, 2007; Rudski, Marra, & Graham, 2004), Spearman's rho correlation coefficient (Hauke & Kossowski, 2011; Lehmann & D'abrera, 1975) was computed to

Table 1
Studies on HGSHS:A Norms

				Age		Overall	Female	Male	
Country	*N*	F/M	Range	*M*	*SD*	Score	Score	Score	Gender Differences
Taiwan	242	137/105	19–30	22.7	2.7	6.87 ± 2.56	7.07 ± 2.40	6.61 ± 2.75	*N.S.*, Cohen's *d* = .18
France uncorrected	115	91/24	18–35	*N.A.*	*N.A.*	8.00 ± 2.47	*N.A.*	*N.A.*	*N.S.*
corrected						6.35 ± 2.98			*N.S.*
Hungary	434	244/190	18–59	26.3	8.05	6.37 ± 2.60	6.63 ± 2.38	6.04 ± 2.83	Cohen's *d* = .23
Portugal	313	247/66	18–59	24.74	7.40	6.73 ± 2.51	6.82 ± 2.43	6.38 ± 2.78	*N.S.*, Cohen's *d* = .18
Poland	1174	968/206	18–53	21.9	4.30	6.26 ± 2.69	5.29 ± 2.58	4.16 ± 2.66	Cohen's *d* = .44
Israel	253	148/104[a]	20–31	24.4	2.35	5.61 ± 2.59	5.4 ± 2.55	5.9 ± 2.64	*N.S.*, Cohen's *d* = −.19
Sweden	291	199/92	15–85	35.4	13.6	6.77 ± 2.50	7.17 ± 2.32	5.91 ± 2.69	Cohen's *d* = .52
Romania	340	218/122	16–38	20.2	3.9	6.24 ± 2.68	*N.A.*	*N.A.*	*N.S.*
Italy	376	297/79	19–26	20.5	2.83	6.41 ± 2.80	6.6 ± 3.02	5.6 ± 2.73	Cohen's *d* = .34
Finland	285	245/40	16–51	21.6	4.2	7.26 ± 2.61	*N.A.*	*N.A.*	*N.S.*
Denmark	376	253/123	19–63	30.0	9.2	7.64 ± 2.50	7.85 ± 2.35	7.20 ± 2.77	Cohen's *d* = .26
Spain	220	153/67	*N.A.*	20.2	2.63	7.13 ± 2.61	*N.A.*	*N.A.*	*N.S.*
Germany	374	192/182	*N.A.*	22.75	*N.A.*	6.51 ± 2.43	*N.A.*	*N.A.*	*N.S.*
Canada	535	357/178	18–47	24.6	*N.A.*	5.38 ± 3.28	*N.A.*	*N.A.*	*N.S.*
Australia	1944	1328/616	*N.A.*	20.5	6.1	5.45 ± 2.95	*N.A.*	*N.A.*	*N.A.*[b]
United States	132	*N.A.*	*N.A.*	*N.A.*	*N.A.*	7.39 ± 3.04	*N.A.*	*N.A.*	*N.A.*

Note. Original norms for the United States sample were set in the United States (Shor & Orne, 1963), followed by the normative data for Australian (Sheehan & McConkey, 1979), Canadian (Montréal; Laurence & Perry, 1982), German (Bongartz, 1985), Spanish (Lamas et al., 1989), Danish (Zachariae et al., 1996), Finnish (Kallio & Ihamuotila, 1999), Italian (De Pascalis, Russo, & Marucci, 2000), Romanian (David, Montgomery, & Holdevici, 2003), Swedish (Bergman et al., 2003), Israeli (Lichtenberg, 2008), Polish (Siuta, 2010), Portuguese (Carvalho, 2013), Hungarian (data only in self-scoring, namely, Hungarian-S; Költő et al., 2015), and French (both including uncorrected and corrected data, namely FRAu and FRAc; Anlló et al., 2017) samples. *N.A.* means that data are not available. *N.S.* means no gender difference in hypnotizability, and Cohen's *d* means the effect size of gender differences.

[a] Lichtenberg (2008) reported that one respondent did not identify gender.

[b] Sheehan and McConkey (1979) calculated the gender differences for Sample 1, Sample 2, and Sample 3 separately but not for aggregate sample, and found that "females scored significantly higher than did males in both Sample 2 and in Sample 3" but not in Sample 1 (p. 297).

examine the relationship between hypnotizability and the female-to-male ratio among 15 reference samples.

Method

Subjects

All subjects gave written informed consent before performing the test. A total of 252 right-handed subjects with normal or corrected-to-normal vision and naive to the HGSHS:A were recruited. All subjects received financial compensation ($5 USD) for their participation in the 70-minute session. The data collection process of the HGSHS:A involved 53 sessions of group tests held between Aug 2014 and May 2016, in a room conditioned specifically for the test and in group sizes that ranged between two and eight people (mode = 4). In addition, 93% of the sample (235 subjects) did not have previous experience with hypnosis. Ten subjects were excluded from further analysis due to the following reasons: two fell asleep during the test administration, six did not finish the questionnaire, and two were foreign students, which left valid results of assessment from 242 Taiwanese subjects (137 females and 105 males; mean age: 22.7 ± 2.7yrs). The study was approved by the Research Ethics Committee of National Taiwan University, Taipei, Taiwan (201311ES018).

Materials

The HGSHS:A presents three standardized phases, including an initial introduction phase to explain and demystify hypnosis, followed by an induction phase to increase relaxation and mental absorption, and finally, 12 suggestions to assess hypnotizability. The induction and suggestion phases can be presented either orally or via audiotape (Carvalho, 2013). The Taiwanese version of the HGSHS:A is an exact translation from the original English version (Shor & Orne, 1962) into Mandarin Chinese. It refers to a simplified Chinese version (Zhou & Wang, 2011) when there was ambiguity in translation. In order to carry out the hypnotic induction in a consistent manner, the verbal instructions to the subjects in the HGSHS:A, including the hypnotic suggestion and 12 test items, were recorded into digital format and played to the subjects under the control of a trained experimenter. The experimenter controlled the audio broadcasting and monitored the progress and safety of the subjects during the session.

Procedures

The administration of the HGSHS:A consisted of three parts, following exactly the instructions of the original version (Shor & Orne, 1962). The procedure begins by having subjects read and sign the consent form. After returning the signed consent form, subjects

received response booklets and were asked not to touch them until the experimenter instructed them to write down their basic personal information (gender and age) on the cover. The experimenter then presented a brief introduction and explanation, closely following the original protocol (Shor & Orne, 1962) to describe the procedure. Some common questions from subjects were also addressed at this point, mainly to clarify stereotypical misconceptions about hypnosis.

Following the introductory phase, the experimenter played the Mandarin Chinese version of the HGSHS:A's hypnotic induction and 12 suggestions, while monitoring the subjects' condition throughout the session. After the recording and following the instructions, subjects completed the response booklet, self-reporting their subjective perception along the dimension specified in each question. The specific instructions of the 12-point scale were presented in the following order: head falling, eye closure, hand lowering, arm immobilization, finger lock, arm rigidity, hands moving, communication inhibition, hallucination, eye catalepsy, posthypnotic suggestion, and amnesia. After all response booklets were completed and returned, the experimenter offered the subjects the opportunity to ask questions and asked them to avoid sharing the participation experience with others. The duration of the initial and final periods was somewhat flexible, depending on the time available for the room reservation and the number of questions raised by the subjects. The recorded hypnotic induction and 12 suggestions took approximately 50 minutes. Thus, the typical HGSHS:A administration session lasted 70 minutes.

Scoring and Data Analysis

The author (ML) scored all booklets based on the subjects' self-report, assigning one point if the suggestion was carried out successfully (Shor & Orne, 1962). Each item was counted only once, no matter how much detail was given. The HGSHS:A's maximum score is 12 points. Note that the scoring of item 12, the amnesia suggestion, followed the reversibility criterion (Kihlstrom & Register, 1984)—namely, one point was assigned only if the subject recalled three or fewer items before receiving the signal to remove the amnesia and two or more additional items after receiving the signal to remember.

The point-biserial item-scale correlations and Kuder-Richardson's KR-20 reliability coefficient were computed. Moreover, the item-total correlations were ranked, and Spearman's rank-order correlations were computed to compare the Taiwanese data with the reference samples. The data processing and statistical analyses were performed using SPSS 20.0 (SPSS, 2011). The level of statistical significance was set at .05.

Results

Score Distribution and Mean Total Scores

Table 2 presents the distribution of the hypnotizability scores. The possible scores range from 0 to 12; however, this sample presented no zeros. Following the criterion suggested in previous studies (Miller, 1975; Nash & Barnier, 2012; Spiegel & Spiegel, 2008), subjects were separated into three categories depending on their level of hypnotizability: 65 subjects were high (scores from 9 to 12; 26.9%), 147 were medium (scores from 4 to 8; 60.6%), and 30 were low (scores from 0 to 3; 12.5%).

The mean HGSHS:A score for the 242 subjects was 6.87 (*SD* = 2.56, SE = .16), which was within the range of the values reported by the reference samples (mean score from 5.38 to 8.00; *SD* from 2.43 to 3.28; SE from .07 to .28). Compared with the reference samples, the Taiwanese mean score was lower than the mean score of the U.S., Spanish, Danish, Finnish, and French uncorrected[1] samples (mean scores from 6.95 to 8.00), but it was higher than the Australian, Canadian, German, Italian, Romanian, Swedish, Israeli, Polish, Portuguese, Hungarian, and French corrected[2] samples (mean scores from 5.38 to 6.77). The score differences by group size are presented in Table 3.

Table 2
HGHS-A Score Distribution in the Taiwan Sample (N = 242)

Score	No. of Cases	% of cases	Cumulative %
0	—	—	—
1	5	2.1	2.1
2	12	5.0	7.1
3	13	5.4	12.5
4	18	7.4	19.9
5	21	8.7	28.6
6	28	11.5	40.1
7	28	11.5	51.6
8	52	21.5	73.1
9	32	13.2	86.3
10	20	8.3	94.6
11	7	2.9	97.5
12	6	2.5	100.0
High (9–12)	65	26.9	26.9
Medium (4–8)	147	60.6	87.5
Low (0–3)	30	12.5	100.0

Note. No score 0.

Score Differences by Group Size

Table 3 shows the distribution of the hypnotizability score by group size. We tested score differences by group size and found significant correlation between group size and the HGSHS:A score ($r = -.189$, $p = .002$) and significant score differences by group size, $F(1, 6) = 4.61$, $p < .001$, partial $\eta^2 = .11$. Table 4 lists the mean scores, standard deviation, and item pass rates for the Taiwanese sample.

Gender Differences

We tested gender differences in hypnotizability and found no significant differences for HGSHS:A scores, $t(240) = 1.38$, $p = .17$ (male: M = 6.61, SD = 2.75, SE = 0.23; female: M = 7.07, SD = 2.40, SE = 0.23).

Item Difficulty

Table 4 also shows the percentages of subjects passing each item for the Taiwanese and the reference samples. The items with the highest pass rates in the Taiwanese sample were eye closure (77%), hands moving (74%), amnesia (73%), and hand lowering (69%); those with the lowest pass rates were hallucination (25%), posthypnotic suggestion (26%), and eye catalepsy (48%). The lowest pass rate was found for item 9, hallucination (25%), which was close to the median percentage of the reference samples. Similar results were reported for the Italian (28%), Finnish (28%), and Spanish (29%) samples in the reference studies. Actually, item pass rates for all suggestions of the Taiwanese sample exceeded neither the lowest nor the highest bounds reported by the reference samples, except for item 12 (amnesia; 73%).

Furthermore, in order to compare the consistency of item difficulty with the reference samples, item pass rates were ranked for each sample

Table 3
HGHS-A Score Distribution by Group Size in Taiwan Sample

Group Size	No. of Sessions	No. of Cases	Mean Score	SE
2	6	12	6.50 ± 2.11	.61
3	11	33	7.36 ± 1.95	.34
4	16	63	7.90 ± 2.45	.31
5	2	10	5.70 ± 3.06	.97
6	2	11	7.45 ± 2.11	.64
7	7	44	5.64 ± 2.89	.44
8	9	69	6.61 ± 2.37	.29

Note. There were total 53 sessions of group test. SE means standard errors.

Table 4
Item Pass Rates (As Percentage), Means, Standard Deviations (SD) and Standard Errors (SE) of the Taiwan and Reference Samples

Country	TWN	FRAu	FRAc	HUN-S	PRT	POL	ISR	SWE	ROM	ITA	FIN	DAN	SPA	GER	CAN	AUS	USA
N	242	115	115	434	313	1174	253	291	340	376	285	376	220	374	535	1944	132
HGSHS:A Item:																	
1. Head falling	50	89	80	68	58	54	48	70	68	70	84	86	73	73	65	61	86
2. Eye closure	77	77	67	67	60	66	78	76	60	62	86	48	64	73	63	57	74
3. Hand lowering	69	91	84	72	68	78	75	66	59	56	89	75	60	83	66	71	89
4. Arm immobilization	64	60	46	41	57	48	37	61	56	55	43	72	58	52	47	36	48
5. Finger lock	57	71	50	71	75	59	50	74	58	60	66	76	67	57	50	53	67
6. Arm rigidity	61	63	48	55	65	58	51	65	59	63	53	75	69	52	47	41	57
7. Hands moving	74	91	83	71	67	71	76	64	61	64	78	78	79	74	64	71	86
8. Communication inhibition	63	75	57	44	51	61	51	56	52	48	56	73	74	49	43	42	50
9. Hallucination	25	18	14	18	12	12	15	14	34	28	28	38	29	47	36	38	56
10. Eye catalepsy	48	69	53	53	46	46	37	51	52	40	52	61	59	47	36	38	56
11. Posthypnotic suggestion	26	42	11	38	44	55	30	15	35	35	37	11	29	31	15	17	36
12. Amnesia	73	52	38	40	72	16	13	65	56	56	53	71	52	36	19	33	48
Mean Percentage per item	57.2	66.7	52.9	53.2	56.3	52.1	46.8	56.4	52.1	53.1	60.4	63.6	59.4	56.1	44.8	45.0	61.3
Sample Mean	6.87	8.00	6.35	6.37	6.73	6.25	5.61	6.77	6.24	6.41	7.26	7.64	7.13	6.51	5.38	5.45	7.39
Sample *SD*	2.56	2.47	2.98	2.60	2.51	2.69	2.59	2.50	2.68	2.80	2.61	2.50	2.61	2.43	3.28	2.95	3.04
Sample SE	0.16	0.23	0.28	0.12	0.14	0.08	0.16	0.15	0.15	0.14	0.15	0.13	0.18	0.13	0.14	0.07	0.26

Note. TWN = Taiwan; FRAu = French, uncorrected; FRAc = French, corrected; HUN-S = Hungarian Self-scores; PRT = Portuguese; POL = Polish; ISR = Israeli; SWE = Swedish; ROM = Romanian; ITA = Italian; FIN = Finnish; DAN = Danish; SPA = Spanish; GER = German; CAN = Canadian; AUS = Australian; USA = United States.

and Spearman correlations (Ramsey, 1989) among all samples were computed. Spearman correlations of the item difficulty for reference samples are presented in Table 5. Spearman correlation coefficients (r_s) between the Taiwanese and other samples ranged from .27 (Danish) to .63 (Portuguese and Finnish). In general, Table 5 shows significant correlations of the Taiwanese sample only with the Portuguese (r_s = .63, p < .05), Finnish (r_s = .63, p < .05), Romanian (r_s = .59, p < .05), and Israeli (r_s = .58, p < .05) samples.

Reliability

The point-biserial item-scale correlations, for measuring how each of the 12 items in the HGSHS:A correlated to the total of the other items, for the Taiwanese and the reference samples (Guilford, 1953; Henrysson, 1963) are reported in Table 6. This measure reflects the relative contribution of an item to the whole scale in reflecting hypnotizability for a group. The coefficients of the point-biserial item-scale correlation in the Taiwanese sample ranged from r = .12 (eye closure and amnesia) to r = .52 (communication inhibition). The item-scale correlation coefficients of hands moving (r = .13) and eye catalepsy (r = .41) in the Taiwanese sample were lower than in any other sample. The correlation coefficients for the other suggestions were within the range of the values in the reference samples.

Table 6 also presents the total scale reliability, measured by the Kuder-Richardson (KR-20) coefficient (Feldt, 1965, 1969; Hoyt, 1941). The magnitude of the total scale reliability coefficient for the Taiwanese sample was .66, which was not much different from the Israeli (.69), Spanish (.68), Swedish (.66), Hungarian (.65), and Portuguese (.63) samples. Across the reference samples, the highest coefficient was found in the Canadian (.84) sample, and the lowest coefficient was found in the German (.62) and Portuguese (.63) samples. In addition, Spearman rank-order correlation coefficients (Ramsey, 1989), were calculated from the ranked item-scale correlations for each sample (see in Table 7) to evaluate the similarity among different samples in terms of rank order of contribution from each item. Spearman rank-order correlation coefficients between the Taiwanese sample and the reference samples, although somewhat lower with the Canadian (r_s = .55), Romanian (r_s = .57), and Israeli (r_s = .57) samples, did not deviate significantly from the mean of such correlation among the 15 reference samples (mean = .51, SD = .06, SE = .02).

Discussion

The current study aimed to establish normative data of the HGSHS: A, a test suitable for examining hypnotic susceptibility in group

Table 5
Spearman Correlations of HGSHS:A Item Difficulty for Taiwan and Reference Samples

Country	FRAu	FRAc	HUN-S	PRT	POL	ISR	SWE	ROM	ITA	FIN	DAN	SPA	GER	CAN	AUS	USA
N	115	115	434	313	1174	253	291	340	376	285	376	220	374	535	1944	132
Taiwan	.54	.51	.44	.63*	.57	.58*	.57	.59*	.53	.63*	.27	.39	.53	.46	.42	.32
N = 242																

Note. TWN = Taiwan; FRAu = French, uncorrected; FRAc = French, corrected; HUN-S = Hungarian Self-scores; PRT = Portuguese; POL = Polish; ISR = Israeli; SWE = Swedish; ROM = Romanian; ITA = Italian; FIN = Finnish; DAN = Danish; SPA = Spanish; GER = German; CAN = Canadian; AUS = Australian; USA = United States.
*$p < .05$.

Table 6
Point-Biserial Item-Scale Correlations and Total Scale Reliability for the Taiwan and Reference Samples

Country	TWN	FRAu	FRAc	HUN-S	PRT	POL	ISR	SWE	ROM	ITA	FIN	DAN	SPA	GER	CAN	AUS	USA
N	242	115	115	434	313	1174	253	291	340	376	285	376	220	374	535	1944	132
HGSHS:A Item:																	
1. Head falling	.24	.27	.37	.26	.29	.38	.19	.36	.30	.29	.20	.17	.30	.21	.44	.39	.34
2. Eye closure	.12	.29	.45	.09	.28	.34	.33	.30	.25	.24	.19	.16	.27	.06	.51	.39	.30
3. Hand lowering	.23	.25	.42	.17	.28	.26	.28	.20	.47	.19	.08	.24	.09	.25	.44	.25	.48
4. Arm immobilization	.40	.34	.39	.38	.31	.36	.31	.34	.56	.29	.48	.45	.38	.33	.53	.36	.66
5. Finger lock	.45	.66	.66	.50	.34	.46	.47	.43	.48	.43	.54	.55	.52	.42	.71	.59	.86
6. Arm rigidity	.47	.41	.50	.53	.43	.47	.52	.42	.57	.35	.41	.44	.51	.42	.70	.55	.89
7. Hands moving	.13	.38	.40	.20	.35	.27	.34	.34	.57	.34	.25	.29	.22	.18	.60	.42	.44
8. Communication inhibition	.52	.84	.70	.47	.43	.49	.53	.45	.55	.38	.40	.44	.40	.38	.65	.51	.78
9. Hallucination	.24	.40	.39	.18	.10	.21	.13	.14	.47	.19	.33	.35	.31	.23	.53	.34	.48
10. Eye catalepsy	.41	.70	.45	.43	.45	.49	.57	.45	.68	.50	.46	.50	.46	.47	.75	.53	.74
11. Posthypnotic suggestion	.23	.05	.36	.11	.03	.12	.02	.07	.26	.30	.27	.19	.11	.14	.47	.18	.46
12. Amnesia	.12	.31	.36	.27	.02	.19	.16	.06	.38	.23	.28	.18	.18	.09	.65	.18	.39
Total Scale Reliability (Kuder-Richardson α)	.66	.70	.80	.65	.63	.70	.69	.66	.71	.70	.71	.70	.68	.62	.84	.76	.80

Note. TWN = Taiwan; FRAu = French, uncorrected; FRAc = French, corrected; HUN-S = Hungarian Self-scores; PRT = Portuguese; POL = Polish; ISR = Israeli; SWE = Swedish; ROM = Romanian; ITA = Italian; FIN = Finnish; DAN = Danish; SPA = Spanish; GER = German; CAN = Canadian; AUS = Australian; USA = United States.

Table 7
Rank Order Correlations (Calculated from Item-Scale Point-Biserial Correlations) for Taiwan and Reference Samples

Country	*N*	FRAu	FRAc	HUN-S	PRT	POL	ISR	SWE	ROM	ITA	FIN	DAN	SPA	GER	CAN	AUS	USA
Taiwan	242	.74**	.59*	.81**	.68*	.74**	.57	.75**	.57	.66*	.76**	.80**	.83**	.88**	.55	.65*	.90**
French, uncorrected	115		.71**	.77**	.75**	.74**	.78**	.74**	.70*	.67*	.73**	.77**	.83**	.72**	.86**	.76**	.73**
French, corrected	115			.53	.74**	.78**	.86**	.77**	.48	.58*	.36	.55	.66*	.64*	.54	.82**	.64*
Hungary	434				.69*	.74**	.67*	.72**	.69*	.68*	.80**	.74**	.83**	.79**	.78**	.71**	.79**
Portugal	313					.91**	.93**	.94**	.82*	.81*	.48*	.66*	.72**	.77*	.60*	.87**	.64*
Poland	1174						.89**	.97**	.60*	.74**	.51	.58*	.82**	.77**	.56*	.75**	.60*
Israel	253							.89**	.72**	.77**	.45	.61*	.69*	.69*	.70*	.87**	.59*
Sweden	291								.63*	.82**	.52	.61*	.78**	.78**	.56	.91**	.60*
Romanian	340									.69*	.60*	.79**	.57	.76**	.68**	.60*	.70**
Italy	376										.62*	.64*	.69*	.64*	.71*	.79**	.59*
Finland	285											.76**	.83*	.72**	.77**	.53	.77*
Denmark	376												.76**	.89**	.72**	.62*	.89**
Spain	220													.78**	.73*	.85**	.74**
Germany	374														.61*	.70*	.91**
Canada	535															.67*	.65*
Australian	1944																.60*
United States	132																–

Note. TWN = Taiwan; FRAu = French, uncorrected; FRAc = French, corrected; HUN-S = Hungarian Self-scores; PRT = Portuguese; POL = Polish; ISR = Israeli; SWE = Swedish; ROM = Romanian; ITA = Italian; FIN = Finnish; DAN = Danish; SPA = Spanish; GER = German; CAN = Canadian; AUS = Australian; USA = United States.
*$p < .05$. **$p < .01$.

administration, for the Mandarin Chinese-speaking population in Taiwan (cf. Roark, Barabasz, Barabasz, and Lin-Roark, 2012, for the SHSS:C [MCT]). In line with HGSHS:A norms previously reported from other countries, our results showed that the mean score, total scale reliability, and most of the item pass rates were within the range of the 15 reference samples. The significant rank-order correlations of item-scale correlations between the Taiwanese sample and 12 other samples indicate that data in the current study are generally congruous with other published norms. In addition, like many other reference samples, we did not observe gender differences, although five other reference samples found female superiority in hypnotizability.

On the other hand, some discrepancies between the current study and the reference norms are worth noting: First, the rank-order correlations of item difficulty are only significant between the Taiwanese sample and four other reference samples. Second, the total scale reliability falls at the lower end of the range of all reference samples. The relatively lower whole-scale reliability and the few significant correlations with other countries in terms of item difficulty warrant closer evaluation.

The relatively lower rank-order correlations in item difficulty between Taiwanese and other samples suggest that the responsivity to each item on the HGSHS:A in Taiwanese sample are dissimilar to those in many of the reference samples. The Taiwanese sample is currently the only one from Asia, which might have culturally or environmentally distinguished it from the reference samples (from North America and Europe). By taking a closer look at the item-by-item scores, item 9 (mosquito hallucination) has the lowest value (25%) and item 12 (amnesia) has the highest value (73%). The low pass rates of the mosquito hallucination can also be found in norms that were acquired between 1999 and 2017, but not before 1989. This may be related to the extent of urbanization: Mosquitos and flies are rare in modern big cities (Becker et al., 2003; Peter, Geiger, Prade, Vogel, & Piesbergen, 2015), which may compromise the ability of mosquito hallucination among city residents, such as the majority of subjects in the Taiwanese sample. In line with this latter "urbanization hypothesis," pass rates on the mosquito hallucination items were above 35% between 1963 (United States) to 1985 (German), yet fell below 30% since 1989 (except for the Danish sample). To summarize, we consider the pass rate of the mosquito hallucination item potentially reflecting the joint influence of climate and urbanization and recommend that the suggestion of hallucination should be adaptive to the local environment of the targeted population.

As for the high pass rates (73%) of the amnesia item, Bergman, Trenter, and Kallio (2003) suspected that misunderstanding the sentence on the HGSHS:A booklet, "Please write down in your

own words a list of all the things that happened since you began looking at the target," might have mistakenly directed subjects to reason what *happened* during the induction and suggestion phases rather than what they had *experienced*. To address the problem, Lamas, Valle-Inclan, Blanco, and Diaz (1989) recommended using the verb "to do" to emphasize what is actually intended ("Please write down in your own words a list of all the things that *you did* since you began looking at the target"). In contrast, Kihlstrom and Register (1984) claimed that item 12 not only involves the suggested amnesia but possibly is subject to the influence from a diversity of factors and suggested that using the reversibility criterion on scoring the amnesia item will considerably lower its passing percentage. At odds with Kihlstrom and Register's findings, when we adopted the reversibility criterion on item 12, the pass rate decreased slightly from 74% to 73%, which differed quite dramatically from the Portuguese sample (which declined from 72% to 29%). It has been suggested that the amnesia item be amended due to its weak and problematic psychometric properties (Woody & Barnier, 2008). As the current study only applied the reversibility criterion but not the accentuated verb "to do," further studies may adjust both the context and scoring of the amnesia item to verify whether better psychometric properties ensue.

The lower KR-20 reliability coefficient of the Taiwanese sample, although still within the range of all reference samples, may be linked to the fact that the more familiar the testing environment is to the subjects, the higher reliability in hypnotizability measures can be observed (Peter, Hagl, Bazijan, & Piesbergen, 2011; Staudacher, Hagl, Piesbergen, & Peter, 2012). Although our subjects were tested at the university campus, testing was not done in a typical classroom. It is likely that the lower reliability resulted from such an unfamiliar environmental setting (Peter et al., 2015). Systematic manipulation of the familiarity with the testing environment would help to clarify the reliability issue in further studies.

Additionally, some notions about hypnotizability are worth consideration: one regarding score differences by group size, and the other regarding gender differences in hypnotizability. For the score differences by group size, Bentler and Roberts (1963) found no difference in hypnotizability scores between small (four to six subjects) and large (39 to 52 subjects) groups. However, we found a significant negative correlation ($r = -.189$, $p = .002$) between the hypnotizability score and size of group testing, indicating that higher hypnotizability would be more frequently assessed in small group (two to four subjects). Our results could not be completely explained by Boucher and Hilgard's (1962) suggestion that subjects with voluntarily

participation scored higher than those participating for class requirement; due to this, we recruited all subjects from student bulletin boards. Therefore, further studies need to control the size of each group session and the volunteer bias in participation to verify the score differences by group size.

Finally, regarding gender differences in hypnotizability, many studies have reported that females surpass males in hypnotizability (Bergman et al., 2003; De Pascalis, Russo, & Marucci, 2000; Költő et al., 2015; Rudski et al., 2004; Siuta, 2010; Zachariae, Sommerlund, & Molay, 1996). Sheehan and McConkey (1979) proposed that the discrepancies in gender differences may be attributed to sample size: Larger differences between the proportion of female and male samples in a study may impact the sensitivity to gender differences. For that reason, we analyzed the correlation between the gender ratio and the mean HGSHS:A score for eight samples that provided information about gender composition, but we did not find significant correlations (r_s = .50, p = .21). In addition, we found that most of the effect sizes for gender differences were small (upper bound approximately .5; see Table 1).

Bem and Lenney (1976) suggested that females and males differ not in hypnotic ability but in hypnotic response; in other words, females tend to make stronger responses when performing the HGSHS:A than males, which may be driven by higher motivation to experience hypnosis or social psychological mechanisms such as sex-role conformist behavior (Költő et al., 2015). Green and Lynn (2010) observed that the three-item expectancy measure is strongly correlated with scores on the HGSHS:A, implying hypnotic expectancy can predict hypnotic responsiveness. They also found that there is a gender difference in the correlation between the HGSHS:A scores and hypnotic expectancy on the good subject item[3]. Therefore, we suspect that gender difference in hypnotizability may be the manifestation of quite diverse components, such as social expectation and personality (Peter et al., 2014). Future work may explore whether hypnotizability can be further deconstructed into hypnotic ability and hypnotic responsiveness and combine the measures of expectation and personality to characterize how individual differences influence hypnotizability.

Despite a few peculiarities in comparison to previous studies, the high significant rank-order correlation of item-scale correlations with other reference samples suggests that the HGSHS:A can effectively evaluate the hypnotizability on the Taiwanese sample, as for many other reference samples. The Mandarin Chinese translation of the HGSHS:A is a reliable and viable instrument for initially screening

subjects' hypnotizability and for further advancing hypnotic research in Taiwan.

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Notes

1. In French uncorrected samples, Anlló, Becchio, and Sackur (2017) followed the scoring criterion as described in the original version (Shor & Orne, 1962) and the criterion of the amnesia in Kihlstrom and Register (1984).
2. In French corrected samples, Anlló et al. (2017) scored by based on Kihlstrom's complementary Scale of Involuntariness (Kihlstrom, 2006).
3. The three-item expectancy measure consists of three questions that asked if subjects thought they would find hypnosis to be interesting, if they would be good subjects, and how many HGSHS:A suggestions they expected to respond to (Green & Lynn, 2010).

Disclosure statement

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Taiwanesische Normen für die Harvard Group Scale of Hypnotic Susceptibility, Form A

MEI-JING LIN UND ERIK CHIHHUNG CHANG

Abstract : Hypnotische Empfänglichkeit ist ein fundamentales individuelles Merkmal, das in Studien zur Untersuchung von Hypnose in Betracht gezogen werden sollte. Obwohl es keine existierenden normativen Daten von Empfänglichkeitstests für Hypnose in Gruppen für die Mandarin-sprechende chinesische Bevölkerung gibt, kam die chinesische Mandarin-Übersetzung der Harvard Group Scale of Susceptibility, Form A

(HGSHS:A) bei 242 Probanden (137 weiblichen und 105 männlichen) in der aktuellen Studie zur Anwendung. Die Ergebnisse weisen darauf hin, daß die Standardeigenschaften, die Verteilungsbreite, die Merkmalquoten und Reliabilität eingeschlossen, mit 15 Referenzproben verglichen werden können. Generell kann die Mandarin-Chinesische Version des HGSHS:A als machbares und verlässliches Instrument verwendet werden, um Teilnehmer in Bezug auf die Hypnotisierbarkeit in der Mandarin-Chinesisch-sprechenden Bevölkerung von Taiwan vorzuscreenen.

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Normes taïwanaises du questionnaire de susceptibilité hypnotique du Groupe de Harvard, formulaire

Résumé: La susceptibilité hypnotique est une caractéristique individuelle fondamentale à prendre en compte dans les études portant sur l'hypnose. Bien qu'il n'existe pas de données normatives portant sur les tests de susceptibilité hypnotique de groupe pour la population chinoise de langue mandarine, les auteurs de cette étude ont administré la traduction en chinois mandarin du questionnaire de susceptibilité hypnotique du Groupe de Harvard, formulaire A (HGSHS:A) à 242 sujets (137 femmes et 105 hommes). Les résultats indiquent que les propriétés normatives, y compris la distribution des scores, les taux de réussite des items et la fiabilité des résultats, sont comparables à ceux de 15 échantillons de référence. De façon générale, la version en chinois mandarin du formulaire HGSHS:A peut être utilisée comme instrument viable et fiable de susceptibilité hypnotique dans la population chinoise de langue mandarine à Taïwan.

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Normas Taiwanesas de la Escala Harvard de Susceptibilidad Hipnótica, Forma A

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Resumen: La susceptibilidad hipnótica es una característica individual fundamental que debe considerarse en estudios que evalúan la hipnosis. Aunque no existen datos normativos de pruebas grupales de susceptibilidad hipnótica para la población China que habla Mandarín, el presente estudio administró la traducción al Mandarín de la Escala Harvard de Susceptibilidad Hipnótica, Forma A (HGSHS:A) a 242 sujetos (137 mujeres y 105 hombres). Los resultados muestran que las propiedades normativas, incluyendo la distribución de puntuaciones, proporción de reactivos aprobados y la fiabilidad son comparables a las 15 muestras de referencia. En general, la versión Mandarín de HGSHS:A puede ser usada como un instrumento viable y fiable para realizar un pre tamizaje de la hipnotizabilidad de sujetos en la población Taiwanesa que habla Chino-Mandarín.

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